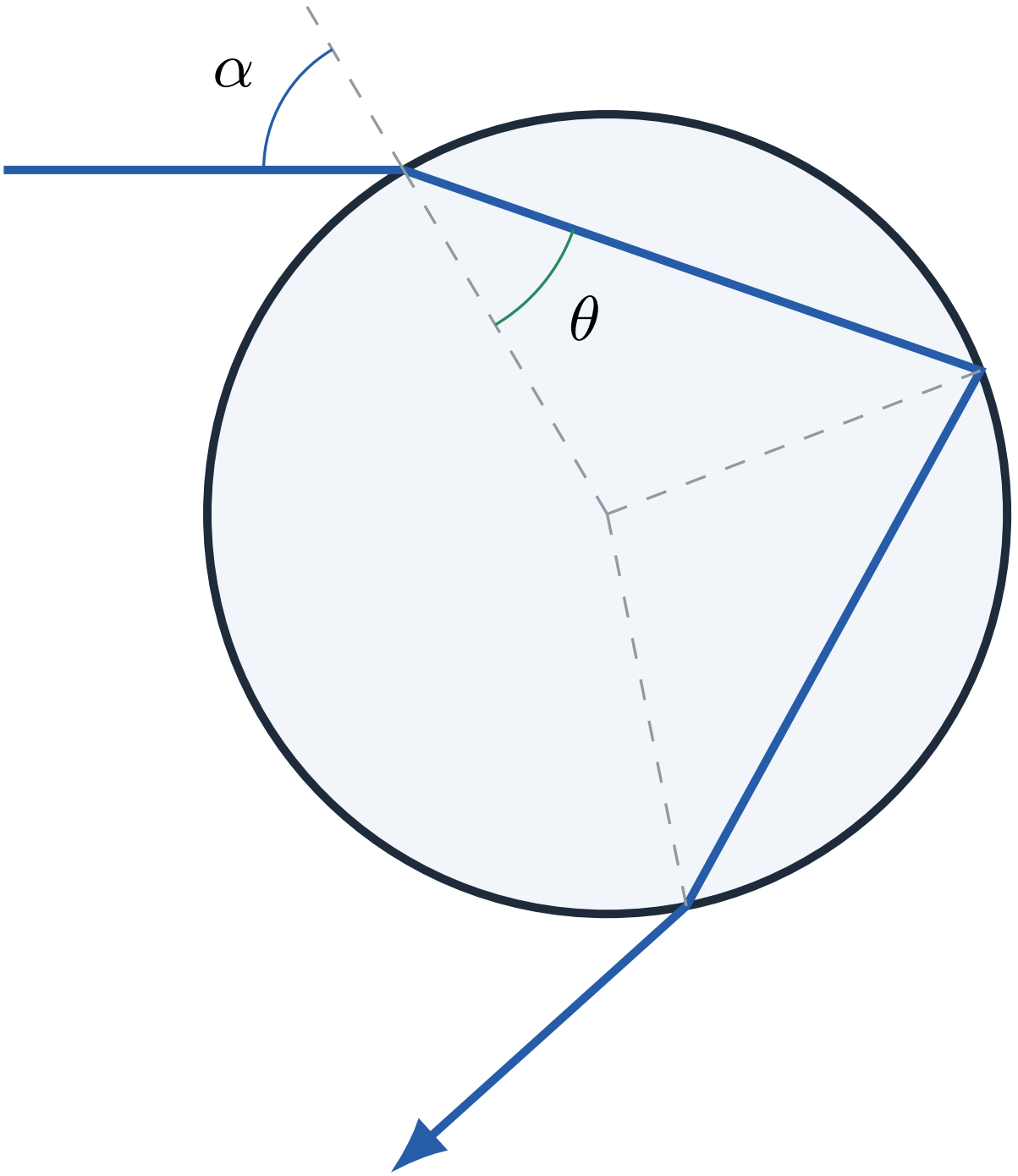
$$\Delta_1 = \pi + 2\alpha - 4\theta;\ \ \alpha_{1m} = 59.39^\circ$$



一级虹：一次内反射，$n = 4/3$